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METHOD AND APPARATUS FOR TRANSMITTING MEASUREMENT CAPABILITY, AND READABLE STORAGE MEDIUM

Abstract

A measurement capability sending method, performed by UE. The measurement capability sending method includes: sending measurement capability indication information to a network device, wherein the measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, and wherein N is an integer greater than 1.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is the U.S. national phase application of International Application No. PCT/CN2022/086350, filed on Apr. 12, 2022, the disclosure of which is incorporated herein by reference in its entirety for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of wireless communication technologies, and in particular to a measurement capability transmission method, a measurement capability transmission apparatus and a readable storage medium.

BACKGROUND

[0003] In a non-terrestrial network (NTN) system, how to make a network device more accurately aware of a terminal capability, especially a measurement capability, of user equipment (UE) is a technical problem to be solved.

SUMMARY

[0004] The present disclosure provides a measurement capability transmission method, a measurement capability transmission apparatus and a readable storage medium.

[0005] In a first aspect, a measurement capability sending method is provided, performed by user equipment (UE) and the measurement capability sending method including: sending measurement capability indication information to a network device, wherein the measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, and wherein N is an integer greater than 1.

[0006] In a second aspect, a measurement capability receiving method is provided, performed by a network device and the measurement capability receiving method including: receiving measurement capability indication information from UE, wherein the measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, and wherein N is an integer greater than 1.

[0007] In a third aspect, an communication device is provided, including one or more processors and one or more memories. The one or more memories are configured to store a computer program, and the one or more processors are configured to execute the computer program and are configured to: send measurement capability indication information to a network device, wherein the measurement capability indication information indicates a measurement capability of whether the communication device supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, and wherein N is an integer greater than 1.

[0008] In the present disclosure, by sending measurement capability indication information to a network device, UE inform a measurement capability of whether to support simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

[0009] It is to be understood that the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The drawings described herein are used to provide further understanding of the present disclosure and constitute a part of the present disclosure. The examples of the present disclosure and their descriptions are used to explain the present disclosure and do not constitute improper limitations on the present disclosure.

[0011] In the drawings, the accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate examples consistent with the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0012] FIG. 1 is a schematic structural diagram of a wireless communication system provided in an example of the present disclosure.

[0013] FIG. 2 illustrates a schematic diagram of a measurement capability transmission method according to an example.

[0014] FIG. 3 illustrates a schematic diagram of a specific illustration according to an example.

[0015] FIG. 4 illustrates a schematic diagram of another specific illustration according to an example.

[0016] FIG. 5 illustrates a schematic diagram of another specific illustration according to an example.

[0017] FIG. 6 illustrates a schematic diagram of another specific illustration according to an example.

[0018] FIG. 7 illustrates a schematic diagram of another specific illustration according to an example.

[0019] FIG. 8 illustrates a schematic diagram of a measurement capability sending method according to an example.

[0020] FIG. 9 illustrates a schematic diagram of a measurement capability sending method according to an example.

[0021] FIG. 10 illustrates a schematic diagram of a measurement capability receiving method according to an example.

[0022] FIG. 11 illustrates a schematic diagram of a measurement capability receiving method according to an example.

[0023] FIG. 12 illustrates a schematic diagram of a measurement capability sending apparatus according to an example.

[0024] FIG. 13 illustrates a schematic diagram of a measurement capability sending device according to an example.

[0025] FIG. 14 illustrates a schematic diagram of a measurement capability receiving apparatus according to an example.

[0026] FIG. 15 illustrates a schematic diagram of a measurement capability receiving device according to an example.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0027] Examples of the present disclosure are further described in conjunction with the accompanying drawings and the detailed description.

[0028] The examples will be described in detail herein, whose illustrations are illustrated in the accompanying drawings. Where the following descriptions involve the drawings, like numerals in different drawings refer to like or similar elements unless otherwise indicated. The implementations described in the following examples do not represent all implementations consistent with the present disclosure. Rather, they are merely examples of apparatuses and methods consistent with some aspects of the present disclosure as detailed in the appended claims.

[0029] The terms used in the examples of the present disclosure are for the purpose of describing

particular examples only, and are not intended to limit the examples of the present disclosure. Terms determined by “a” and “the” in their singular forms used in the examples of the present disclosure and the appended claims are also intended to include their plural forms, unless clearly indicated otherwise in the context. It is also to be understood that the term “and/or” as used herein is and includes any and all possible combinations of one or more of the associated listed items.

[0030] It is to be understood that, although terms “first,” “second,” “third,” and the like may be adopted in the examples of the present disclosure to describe various information, such information should not be limited to these terms. These terms are only used to distinguish the information of the same type with each other. For example, without departing from the scope of the examples of the present disclosure, first information may be referred as second information; and similarly, second information may also be referred as first information. Depending on the context, the word “if” as used herein may be interpreted as “when,” “upon,” or “in response to determining.”

[0031] The following describes in detail the examples of the present disclosure. Illustrations of the examples are illustrated in the accompanying drawings, with the same or similar reference numerals referring to the same or similar elements throughout. The examples, which are described below with reference to the accompanying drawings, are illustrated and are intended to explain the present disclosure, but should not be construed as a limitation of the present disclosure.

[0032] As illustrated in FIG. 1, a measurement capability transmission method provided in an example of the present disclosure may be applied to a wireless communication system **100**. The wireless communication system **100** may include, but is not limited to, a network device **101** and user equipment (UE) **102**. The UE **102** is configured to support carrier aggregation. The UE **102** may connect to multiple carrier units of the network device **101**, including a primary carrier unit and one or more secondary carrier units.

[0033] It is to be understood that the wireless communication system **100** is applicable to both low-frequency scenarios and high-frequency scenarios. Application scenarios of the wireless communication system **100** include, but are not limited to, a long term evolution (LTE) system, a LTE frequency division duplex (FDD) system, a LTE time division duplex (TDD) system, a worldwide interoperability for micro wave access (WiMAX) communication system, a cloud radio access network (CRAN) system, a future fifth-generation (5G) system, a new radio (NR) communication system or a future evolved public land mobile network (PLMN) system, etc.

[0034] The UE **102** illustrated above may be user equipment, a terminal, an access terminal, a terminal unit, a terminal station, a mobile station (MS), a remote station, a remote terminal, a mobile terminal, a wireless communication device, a terminal agent or a user device, etc. The UE **102** may be equipped with a wireless transceiving function, and may perform communications (such as wireless communications) with one or more network devices **101** of one or more communication systems, and receive network services provided by the one or more network devices **101**. The network device **101** here includes, but is not limited to, the base station illustrated in the drawing.

[0035] The UE **102** may be a cellular phone, a cordless phone, a session initiation protocol (SIP) phone, a wireless local loop (WLL) station, a personal digital assistant (PDA) device, a handheld device with a wireless communication function, a computing device or another processing device connected to a wireless modem, a vehicle-mounted device, a wearable device, UE in a future 5G network, or UE in a future evolved PLMN network, etc.

[0036] The network device **101** may be an access network device (or called an access network node). The access network device refers to a device that provides a network access function, for example, a radio access network (RAN) base station, etc. The network device may specifically include a base station (BS) device, or include the BS device and a radio resource management device for controlling the BS device, etc. The network device may also include a relay station (a relay device), an access point, a BS in the future 5G network, a BS in the future evolved PLMN network, or a NR BS, etc. The network device may be a wearable device or a vehicle-mounted

device. The network device may be a communication chip having a communication module.

[0037] For example, the network device **101** includes, but is not limited to, a next-generation gnodeB (gNB) in 5G, an evolved node B (eNB) in an LTE system, a radio network controller (RNC), a node B (NB) in a WCDMA system, a radio controller under a CRAN system, a BS controller (BSC), a base transceiver station (BTS) in a GSM system or a CDMA system, a home BS (for example, a home evolved nodeB or a home node B (HNB)), a baseband unit (BBU), a transmitting and receiving point (TRP), a transmitting point (TP) or a mobile switching center, etc.

[0038] An example of the present disclosure provides a measurement capability transmission method. FIG. 2 illustrates a flowchart of a measurement capability transmission method according to an example. As illustrated in FIG. 2, the method includes steps **S201-S202**.

[0039] At step **S201**, UE sends measurement capability indication information to a network device.

[0040] The measurement capability indication information sent from the UE to the network device indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, where N is an integer greater than 1.

[0041] In some possible implementations, the same measurement window is a synchronization signal block (SSB)-based radio resource management (RRM) measurement timing configuration window (SMTC) used by the UE when performing the neighbor cell measurement.

[0042] In some possible implementations, the same measurement window is a channel state information reference signal (CSI-RS) based measurement timing configuration window (CMTC) used by the UE when performing the neighbor cell measurement.

[0043] In some possible implementations, any two of the N different satellites correspond to different satellite types. The different satellite types correspond to different satellite orbit altitude ranges.

[0044] The satellite types includes at least one of the following types: a low earth orbit (LEO), which is a near-circular orbit about 200 to 1,200 kilometers from the ground; a medium earth orbit (MEO), which is a circular orbit about 1,200 to 36,000 kilometers from the ground; a geosynchronous orbit (GEO), also known as a high earth orbit (HEO), which is a circular orbit about 36,000 kilometers from the ground; a geostationary orbit (GSO), which is a geosynchronous orbit about 36,000 kilometers from the ground and with an angle of 0° between the orbital plane and the equatorial plane; or a non-geostationary orbit (NGSO), which is an orbit that is not synchronized with the earth.

[0045] The satellite types may include only the LEO, the MEO or the GEO, or include only the GSO or the NGSO, or include a combination of two types. For example, the satellite type of one satellite is both NGSO and LEO, or the satellite type of one satellite is both NGSO and MEO.

[0046] In an illustration as illustrated in FIG. 3, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell B **32** covered by Satellite 2 **35** of the LEO-type, and Cell C **33** covered by Satellite 3 **36** of the MEO-type.

[0047] The UE may simultaneously measure Cell B **32** and Cell C **33** of different satellite types within the same measurement window.

[0048] In an illustration as illustrated in FIG. 4, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell B **32** covered by Satellite 2 **35** of the NGSO-type, and Cell C **33** covered by Satellite 3 **36** of the GSO-type.

[0049] The UE may simultaneously measure Cell B **32** and Cell C **33** of different satellite types within the same measurement window.

[0050] In an illustration as illustrated in FIG. 5, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell D **54** covered by Satellite 4 **51** of the LEO-type, Cell E **55** covered by Satellite 5 **52** of the MEO-type, and Cell F **56** covered by Satellite 6 **53** of the GEO-type.

[0051] The UE may simultaneously measure Cell D **54**, Cell E **55** and Cell F **56** of different

satellite types within the same measurement window.

[0052] In some possible implementations, at least two of the N different satellites correspond to different satellite types, with different satellite types corresponding to different satellite orbit altitude ranges.

[0053] In an illustration as illustrated in FIG. 6, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell D **54** covered by Satellite 4 **51** of the LEO-type, Cell E **55** covered by Satellite 5 **52** of the LEO-type, and Cell F **56** covered by Satellite 6 **53** of the GEO-type.

[0054] The UE may simultaneously measure Cell D **54**, Cell E **55** and Cell F **56** within the same measurement window.

[0055] Since different neighbor cells correspond to different satellite orbits, the terminal may not receive signals from the satellites in different orbits at the same time, and may not measure the satellites in different orbits within the same measurement window. In this case, it is to measure the satellites in different orbits in sequence, resulting in different measurement delays and different generated measurement reports. In these implementations, the UE sends to the network device the measurement capability of whether it supports the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window when performing the neighbor cell measurement, and at least two of the N different satellites correspond to different satellite types, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

[0056] In some possible implementations, the N different satellites correspond to the same satellite type, with different satellite types corresponding to different satellite orbit altitude ranges.

[0057] In some possible implementations, at least two of the N different satellites correspond to the same satellite types, with different satellite types corresponding to different satellite orbit altitude ranges.

[0058] In an illustration as illustrated in FIG. 7, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell G **71** covered by Satellite 8 **72** of the LEO-type, and Cell H **73** covered by Satellite 9 **74** of the LEO-type.

[0059] The UE may simultaneously measure Cell G **71** and Cell H **72** of the same satellite type within the same measurement window.

[0060] In some possible implementations, the cells corresponding to the N different satellites are intra-frequency cells. Alternatively, the cells corresponding to at least two of the N different satellites are inter-frequency cells.

[0061] At step S**202**, the UE sends a neighbor cell measurement report to the network device.

[0062] The neighbor cell measurement report sent from the UE to the network device includes measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.

[0063] In the examples of the present disclosure, by sending the measurement capability indication information to the network device, the UE informs the measurement capability of whether to support the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window when performing the neighbor cell measurement, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

[0064] An example of the present disclosure provides a measurement capability sending method, which is performed by UE. FIG. 8 illustrates a flowchart of a measurement capability sending method according to an example. As illustrated in FIG. 8, the method includes the following step.

[0065] At step S**801**, the UE sends measurement capability indication information to a network device.

[0066] The measurement capability indication information sent from the UE to the network device indicates a measurement capability of whether the UE supports simultaneous measurements on

cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, where N is an integer greater than 1.

[0067] In some possible implementations, the same measurement window is an SSB-based RRM measurement timing configuration window (SMTC) used by the UE when performing the neighbor cell measurement.

[0068] In some possible implementations, the same measurement window is a CSI-RS based measurement timing configuration window (CMTC) used by the UE when performing the neighbor cell measurement.

[0069] In some possible implementations, any two of the N different satellites correspond to different satellite types, with the different satellite types corresponding to different satellite orbit altitude ranges.

[0070] The satellite types include at least one of the following types: an LEO, which is a near-circular orbit about 200 to 1,200 kilometers from the ground; an MEO, which is a circular orbit about 1,200 to 36,000 kilometers from the ground; a GEO, also known as an HEO, which is a circular orbit about 36,000 kilometers from the ground; a GSO, which is a geosynchronous orbit about 36,000 kilometers from the ground and with an angle of 0° between the orbital plane and the equatorial plane; or an NGSO, which is an orbit that is not synchronized with the earth.

[0071] The satellite types may include only the LEO, the MEO or the GEO, or include only the GSO or the NGSO, or include a combination of two types. For example, the satellite type of one satellite is both NGSO and LEO, or the satellite type of one satellite are both NGSO and MEO.

[0072] In an illustration as illustrated in FIG. 3, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell B **32** covered by Satellite 2 **35** of the LEO-type, and Cell C **33** covered by Satellite 3 **36** of the MEO-type.

[0073] The UE may simultaneously measure Cell B **32** and Cell C **33** of different satellite types within the same measurement window.

[0074] In an illustration as illustrated in FIG. 4, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell B **32** covered by Satellite 2 **35** of the NGSO-type, and Cell C **33** covered by Satellite 3 **36** of the GSO-type.

[0075] The UE may simultaneously measure Cell B **32** and Cell C **33** of different satellite types within the same measurement window.

[0076] In an illustration as illustrated in FIG. 5, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell D **54** covered by Satellite 4 **51** of the LEO-type, Cell E **55** covered by Satellite 5 **52** of the MEO-type, and Cell F **56** covered by Satellite 6 **53** of the GEO-type.

[0077] The UE may simultaneously measure Cell D **54**, Cell E **55** and Cell F **56** of different satellite types within the same measurement window.

[0078] In some possible implementations, at least two of the N different satellites correspond to different satellite types, with different satellite types corresponding to different satellite orbit altitude ranges.

[0079] In an illustration as illustrated in FIG. 6, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell D **54** covered by Satellite 4 **51** of the LEO-type, Cell E **55** covered by Satellite 5 **52** of the LEO-type, and Cell F **56** covered by Satellite 6 **53** of the GEO-type.

[0080] The UE may simultaneously measure Cell D **54**, Cell E **55** and Cell F **56** within the same measurement window.

[0081] Since different neighbor cells correspond to different satellite orbits, the terminal may not receive signals from the satellites in different orbits at the same time, and may not measure the satellites in different orbits within the same measurement window. In this case, it is to measure the satellites in different orbits in sequence, resulting in different measurement delays and different generated measurement reports. In these implementations, the UE sends to the network device the

measurement capability of whether it supports the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window when performing the neighbor cell measurement, and at least two of the N different satellites correspond to different satellite types, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

[0082] In some possible implementations, the N different satellites correspond to the same satellite type, with different satellite types corresponding to different satellite orbit altitude ranges.

[0083] In some possible implementations, at least two of the N different satellites correspond to the same satellite types, with different satellite types corresponding to different satellite orbit altitude ranges.

[0084] In an illustration as illustrated in FIG. 7, the UE resides in Cell A **31** covered by Satellite 1 **34** of the GEO-type, and its neighbor cells include: Cell G **71** covered by Satellite 8 **72** of the LEO-type, and Cell H **73** covered by Satellite 9 **74** of the LEO-type.

[0085] The UE may simultaneously measure Cell G **71** and Cell H **72** of the same satellite type within the same measurement window.

[0086] In some possible implementations, the cells corresponding to the N different satellites are intra-frequency cells. Alternatively, the cells corresponding to at least two of the N different satellites are inter-frequency cells.

[0087] In the examples of the present disclosure, by sending the measurement capability indication information to the network device, the UE informs the measurement capability of whether to support the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window when performing the neighbor cell measurement, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

[0088] An example of the present disclosure provides a measurement capability sending method, which is performed by UE. FIG. 9 illustrates a flowchart of a measurement capability sending method according to an example. As illustrated in FIG. 9, the method includes the following steps.

[0089] At step **S901**, measurement capability indication information is sent to a network device.

[0090] The measurement capability indication information sent by the UE indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, where N is an integer greater than 1.

[0091] At step **S902**, a neighbor cell measurement report is sent to the network device.

[0092] The neighbor cell measurement report sent from the UE to the network device includes measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.

[0093] An example of the present disclosure provides a measurement capability receiving method, which is performed by a network device. FIG. 10 illustrates a flowchart of a measurement capability receiving method according to an example. As illustrated in FIG. 10, the method includes the following step.

[0094] At step **S1001**, measurement capability indication information sent by UE is received.

[0095] The received measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, where N is an integer greater than 1.

[0096] In some possible implementations, the same measurement window is an SSB-based RRM measurement timing configuration window (SMTC) used by the UE when performing the neighbor cell measurement.

[0097] In some possible implementations, the same measurement window is a CSI-RS based measurement timing configuration window (CMTC) used by the UE when performing the neighbor

cell measurement.

[0098] In some possible implementations, any two of the N different satellites correspond to different satellite types, with the different satellite types corresponding to different satellite orbit altitude ranges.

[0099] The satellite types include at least one of the following types: an LEO, which is a near-circular orbit about 200 to 1,200 kilometers from the ground; an MEO, which is a circular orbit about 1,200 to 36,000 kilometers from the ground; a GEO, also known as an HEO, which is a circular orbit about 36,000 kilometers from the ground; a GSO, which is a geosynchronous orbit about 36,000 kilometers from the ground and with an angle of 0° between the orbital plane and the equatorial plane; or an NGSO, which is an orbit that is not synchronized with the earth.

[0100] The satellite types may include only the LEO, the MEO or the GEO, or include only the GSO or the NGSO, or include a combination of two types. For example, the satellite type of one satellite is both NGSO and LEO, or the satellite type of one satellite are both NGSO and MEO.

[0101] The illustrations illustrated as FIG. 3, FIG. 4 and FIG. 5 have the same descriptions as the above examples, which are not repeated here.

[0102] In some possible implementations, at least two of the N different satellites correspond to different satellite types, with different satellite types corresponding to different satellite orbit altitude ranges.

[0103] The illustration illustrated as FIG. 6 has the same descriptions as the above example, which is not repeated here.

[0104] Since different neighbor cells correspond to different satellite orbits, the terminal may not receive signals from the satellites in different orbits at the same time, and may not measure the satellites in different orbits within the same measurement window. In this case, it is to measure the satellites in different orbits in sequence, resulting in different measurement delays and different generated measurement reports. In these implementations, the UE sends to the network device the measurement capability of whether it supports the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window when performing the neighbor cell measurement, and at least two of the N different satellites correspond to different satellite types, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

[0105] In some possible implementations, the N different satellites correspond to the same satellite type, with different satellite types corresponding to different satellite orbit altitude ranges.

[0106] In some possible implementations, at least two of the N different satellites correspond to the same satellite type, with different satellite types corresponding to different satellite orbit altitude ranges.

[0107] The illustration illustrated as FIG. 7 has the same descriptions as the above example, which is not repeated here.

[0108] In the example of the present disclosure, by receiving the measurement capability indication information sent by the UE, the network device is informed the measurement capability of whether the UE supports the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window when performing the neighbor cell measurement, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

[0109] An example of the present disclosure provides a measurement capability receiving method, which is performed by a network device. FIG. 11 illustrates a flowchart of a measurement capability receiving method according to an example. As illustrated in FIG. 11, the method includes the following steps.

[0110] At step S1101, measurement capability indication information sent by UE is received.

[0111] The received measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N

different satellites within a same measurement window when performing a neighbor cell measurement, where N is an integer greater than 1.

[0112] At step **S1102**, a neighbor cell measurement report sent by the UE is received.

[0113] The neighbor cell measurement report sent from the UE to the network device includes measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.

[0114] Based on the same concept as the method examples, the present disclosure also provides an communication apparatus. The communication apparatus may be equipped with the functions of the UE **102** in the method examples and is configured to perform the step(s) performed by the UE **102** provided in the above examples. The functions may be implemented by hardware, or by corresponding software executed by hardware. The hardware or software includes one or more modules corresponding to the above functions.

[0115] In a possible implementation, the communication apparatus **1200** as illustrated in FIG. **12** may serve as the UE **102** involved in the above method examples, and perform the step(s) performed by the UE **102** in a method example.

[0116] The communication apparatus **1200** includes a transceiving module **1201**, which is configured to send measurement capability indication information to a network device. The measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, where N is an integer greater than 1.

[0117] The transceiving module **1201** is further configured to send a neighbor cell measurement report to the network device. The neighbor cell measurement report includes measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.

[0118] In some possible implementations, at least two of the N different satellites correspond to different satellite types, with different satellite types corresponding to different satellite orbit altitude ranges.

[0119] In some possible implementations, the cells corresponding to the N different satellites are intra-frequency cells. Alternatively, the cells corresponding to at least two of the N different satellites are inter-frequency cells.

[0120] When being the UE **102**, the communication apparatus may further be provided as the device as illustrated in FIG. **13**.

[0121] Referring to FIG. **13**, the device **1300** may include one or more of the following components: a processing component **1302**, a memory **1304**, a power supply component **1306**, a multimedia component **1308**, an audio component **1310**, an input/output (I/O) interface **1312**, a sensor component **1314**, and a communication component **1316**.

[0122] The processing component **1302** generally controls the overall operations of the device **1300**, such as operations associated with display, phone calls, data communications, camera operations, and recording operations. The processing component **1302** may include one or more processors **1320** to execute instructions to complete all or a part of the steps of the above methods. In addition, the processing component **1302** may include one or more modules which facilitate the interaction between the processing component **1302** and other components. As an example, the processing component **1302** may include a multimedia module to facilitate the interaction between the multimedia component **1308** and the processing component **1302**.

[0123] The memory **1304** is configured to store various types of data to support the operations of the device **1300**. Examples of such data include instructions for any application or method operated on the device **1300**, contact data, phonebook data, messages, pictures, videos, and the like. The memory **1304** may be implemented by any type of volatile or non-volatile storage device or a combination thereof, such as a static random access memory (SRAM), an electrically erasable

programmable read-only memory (EEPROM), an erasable and programmable read-only memory (EPROM), a programmable read-only memory (PROM), a read-only memory (ROM), a magnetic memory, a flash memory, a magnetic disk or an optical disk.

[0124] The power supply component **1306** provides power for various components of the device **1300**. The power supply component **1306** may include a power management system, one or more power supplies, and other components associated with generating, managing, and distributing power for the device **1300**.

[0125] The multimedia component **1308** includes a screen providing an output interface between the device **1300** and a user. In some examples, the screen may include a liquid crystal display (LCD) and a touch panel (TP). If the screen includes the TP, the screen may be implemented as a touch screen to receive input signals from the user. The TP may include one or more touch sensors to sense touches, swipes, and gestures on the TP. The touch sensors may not only sense a boundary of a touch or swipe, but also sense a lasting time and a pressure associated with the touch or swipe. In some examples, the multimedia component **1308** includes a front camera and/or a rear camera. The front camera and/or rear camera may receive external multimedia data when the device **1300** is in an operating mode, such as a photographing mode or a video mode. Each front camera and rear camera may be a fixed optical lens system or have focal length and optical zooming capability.

[0126] The audio component **1310** is configured to output and/or input audio signals. For example, the audio component **1310** includes a microphone (MIC) that is configured to receive an external audio signal when the device **1300** is in an operating mode, such as a call mode, a recording mode, and a voice recognition mode. The received audio signal may be further stored in memory **1304** or transmitted via communication component **1316**. In some examples, the audio component **1310** also includes a speaker for outputting audio signals.

[0127] The I/O interface **1312** provides an interface between the processing component **1302** and a peripheral interface module. The above peripheral interface module may be a keyboard, a click wheel, buttons, or the like. These buttons may include but not limited to a home button, a volume button, a start button and a lock button.

[0128] The sensor component **1314** includes one or more sensors to provide the device **1300** with status assessments in various aspects. For example, the sensor component **1314** may detect an open/closed state of the device **1300** and a relative positioning of components such as the display and keypad of the device **1300**, and the sensor component **1314** may also detect a change in position of the device **1300** or a component of the device **1300**, the presence or absence of user contact with the device **1300**, orientation or acceleration/deceleration of the device **1300**, and temperature change of the device **1300**. The sensor component **1314** may include a proximity sensor configured to detect the presence of a nearby object without any physical contact. The sensor component **1314** may also include a light sensor, such as a CMOS or CCD image sensor, for being applied in imaging applications. In some examples, the sensor component **1314** may also include an acceleration sensor, a gyro sensor, a magnetic sensor, a pressure sensor, or a temperature sensor.

[0129] The communication component **1316** is configured to facilitate wired or wireless communication between the device **1300** and other devices. The device **1300** may access a wireless network based on a communication standard, such as Wi-Fi, 4G, 5G, or a combination thereof. In an example, the communication component **1316** receives broadcast signals or broadcast related information from an external broadcast management system via a broadcast channel. In an example, the communication component **1316** also includes a near field communication (NFC) module to facilitate short-range communication. For example, the NFC module may be implemented based on a radio frequency identification (RFID) technology, an infrared data association (IrDA) technology, an ultra-wideband (UWB) technology, a Bluetooth® (BT) technology and other technologies.

[0130] In one or more examples, the device **1300** may be implemented by at least one application

specific integrated circuit (ASIC), digital signal processor (DSP), digital signal processing device (DSPD), programmable logic device (PLD), field programmable gate array (FPGA), controller, microcontroller, microprocessor, or another electronic component for performing the above methods.

[0131] In one or more examples, there is also provided a non-transitory computer-readable storage medium including instructions, such as the memory **1304** including instructions. These instructions may be executed by the one or more processors **1320** of the device **1300** to complete the above methods. For example, the non-transitory computer-readable storage medium may be a read-only memory (ROM), a random access memory (RAM), a CD-ROM, a magnetic tape, a floppy disk, an optical data storage device, and the like.

[0132] Based on the same concept as the method examples, the present disclosure also provides a communication apparatus. The communication apparatus may be equipped with the functions of the network device **101** in the method examples and is configured to perform the step(s) performed by the network device **101** provided in the above examples. The functions may be implemented by hardware, or by corresponding software executed by hardware. The hardware or software includes one or more modules corresponding to the above functions.

[0133] In a possible implementation, the communication apparatus **1400** as illustrated in FIG. **14** may serve as the network device **101** involved in the above method examples, and perform the step(s) performed by the network device **101** in the method examples.

[0134] The communication apparatus **1400** as illustrated in FIG. **14** includes a transceiving module **1401**.

[0135] The transceiving module **1401** is configured to receive measurement capability indication information from UE. The measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, where N is an integer greater than 1.

[0136] In some possible implementations, the transceiving module **1401** is further configured to: receive a neighbor cell measurement report from the UE. The neighbor cell measurement report includes measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.

[0137] In some possible implementations, at least two of the N different satellites correspond to different satellite types, with different satellite types corresponding to different satellite orbit altitude ranges.

[0138] In some possible implementations, the cells corresponding to the N different satellites are intra-frequency cells. Alternatively, the cells corresponding to at least two of the N different satellites are inter-frequency cells.

[0139] When being a network device, the structure of a communication device may further be as illustrated in FIG. **15**. The structure of the communication device is described by taking the network device **101** as a base station as an example. As illustrated in FIG. **15**, the device **1500** includes a memory **1501**, a processor **1502**, a transceiving component **1503**, and a power supply component **1506**. The memory **1501** is coupled to the processor **1502** and may be configured to store programs and data required by the communication device **1500** to implement various functions. The processor **1502** is configured to support the communication device **1500** to perform the corresponding functions in the above methods. These functions may be implemented by calling the programs stored in the memory **1501**. The transceiving component **1503** may be a wireless transceiver, which may be configured to support the communication device **1500** to receive signaling and/or data and send signaling and/or data through a wireless air interface. The transceiving component **1503** may also be referred to as a transceiving unit or a communication unit. The transceiving component **1503** may include a radio frequency component **1504** and one or more antennas **1505**. The radio frequency component **1504** may be a remote radio-frequency unit

(RRU), which may be specifically configured to transmit radio frequency signals and convert between radio frequency signals and baseband signals. The one or more antennas **1505** may be specifically configured to radiate and receive radio frequency signals.

[0140] When the communication device **1500** is to send data, the processor **1502** may perform a baseband processing on the data to be sent and output the baseband signals to a radio-frequency unit. The radio-frequency unit performs a radio-frequency processing on the baseband signals and then sends the radio-frequency signals in the form of electromagnetic waves through the one or more antennas. When data is sent to the communication device **1500**, the radio-frequency unit receives the radio-frequency signals through the one or more antennas, converts the radio-frequency signals into baseband signals, and outputs the baseband signals to the processor **1502**. The processor **1502** converts the baseband signals into data and processes the data.

[0141] Other implementations of the examples of the present disclosure will be readily apparent to those skilled in the art after implementing the present disclosure by referring to the description. The present disclosure is intended to cover any variations, uses, or adaptations of the present disclosure that are in accordance with the general principles thereof and include common general knowledge or conventional technical means in the art that are not disclosed in the present disclosure. The description and the examples are only illustrative, and the scope and spirit of the examples of the present disclosure are to be indicated by appended claims.

[0142] It is to be understood that the examples of the present disclosure is not limited to the precise structures described above and illustrated in the accompanying drawings, and that various modifications and changes may be made without departing from the scope thereof. The scope of the examples of the present disclosure is limited only by the appended claims.

INDUSTRIAL PRACTICALITY

[0143] By sending measurement capability indication information to a network device, the UE informs a measurement capability of whether to support simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, so that the network device accurately learns the measurement capability of the UE, and thereby applies the measurement capability to appropriate scheduling processing.

Claims

1. A measurement capability sending method, performed by user equipment (UE), the measurement capability sending method comprising: sending measurement capability indication information to a network device, wherein the measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, and wherein N is an integer greater than 1.
2. The method according to claim 1, further comprising: sending a neighbor cell measurement report to the network device, wherein the neighbor cell measurement report comprises measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.
3. The method according to claim 1, wherein at least two of the N different satellites correspond to different satellite types, and wherein different satellite types correspond to different satellite orbit altitude ranges.
4. The method according to claim 1, wherein the cells corresponding to the N different satellites are intra-frequency cells, or the cells corresponding to at least two of the N different satellites are inter-frequency cells.
5. A measurement capability receiving method, performed by a network device, the measurement capability receiving method comprising: receiving measurement capability indication information

from user equipment (UE), wherein the measurement capability indication information indicates a measurement capability of whether the UE supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, and wherein N is an integer greater than 1.

6. The method according to claim 5, further comprising: receiving a neighbor cell measurement report from the UE, wherein the neighbor cell measurement report comprises measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.

7. The method according to claim 5, wherein at least two of the N different satellites correspond to different satellite types, and wherein different satellite types correspond to different satellite orbit altitude ranges.

8. The method according to claim 5, wherein the cells corresponding to the N different satellites are intra-frequency cells, or the cells corresponding to at least two of the N different satellites are inter-frequency cells.

9.-10. (canceled)

11. A communication device, comprising one or more processors, and one or more memories, wherein the one or more memories are configured to store a computer program; and wherein the one or more processors are configured to execute the computer program and are configured to: send measurement capability indication information to a network device, wherein the measurement capability indication information indicates a measurement capability of whether the communication device supports simultaneous measurements on cells corresponding to N different satellites within a same measurement window when performing a neighbor cell measurement, and wherein N is an integer greater than 1.

12. A communication device, comprising one or more processors, and one or more memories, wherein the one or more memories are configured to store a computer program; and wherein the one or more processors are configured to execute the computer program to implement the method according to claim 5.

13. A non-transitory computer-readable storage medium, in which instructions are stored, wherein the instructions, when called and executed on a computer, enable the computer to perform the method according to claim 1.

14. A non-transitory computer-readable storage medium, in which instructions are stored, wherein the instructions, when called and executed on a computer, enable the computer to perform the method according to claim 5.

15. The method according to claim 2, wherein at least two of the N different satellites correspond to different satellite types, and wherein different satellite types correspond to different satellite orbit altitude ranges.

16. The method according to claim 2, wherein the cells corresponding to the N different satellites are intra-frequency cells, or the cells corresponding to at least two of the N different satellites are inter-frequency cells.

17. The method according to claim 6, wherein at least two of the N different satellites correspond to different satellite types, and wherein different satellite types correspond to different satellite orbit altitude ranges.

18. The method according to claim 6, wherein the cells corresponding to the N different satellites are intra-frequency cells, or the cells corresponding to at least two of the N different satellites are inter-frequency cells.

19. The communication device according to claim 11, wherein the one or more processors are further configured to: send a neighbor cell measurement report to the network device, wherein the neighbor cell measurement report comprises measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.

- 20.** The communication device according to claim 11, wherein at least two of the N different satellites correspond to different satellite types, and wherein different satellite types correspond to different satellite orbit altitude ranges.
- 21.** The communication device according to claim 11, wherein the cells corresponding to the N different satellites are intra-frequency cells, or the cells corresponding to at least two of the N different satellites are inter-frequency cells.
- 22.** The communication device according to claim 12, wherein the one or more processors are further configured to: receive a neighbor cell measurement report from the UE, wherein the neighbor cell measurement report comprises measurement results of the simultaneous measurements on the cells corresponding to the N different satellites within the same measurement window.
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